Historical Simulation VaR

Independent validation report | Sai Vakkalagadda | FIM 500 | October 23, 2026

1. Scope and preliminary assessment

The equal-weighted Historical Simulation model is reproducible on the frozen team dataset, but its 99% forecasts underestimate the frequency of tail losses and its exceptions are dependent over time. There are 59 exceptions in 3,939 forecasts (1.50%), versus 39.39 expected. Both the Kupiec coverage and Christoffersen independence null hypotheses are rejected at 5%. These are preliminary findings, not a final model effectiveness rating.

Specification	Baseline
$1M exposure	Constant long SPY exposure; daily P&L = $1,000,000 x simple return
Risk measure	1-day, 99% VaR; positive dollar loss threshold
Data	Bloomberg Last Price; unadjusted, price-only; Jan 2010-Aug 31, 2026
Estimation	250 prior returns; first forecast Dec 31, 2010
Evaluation	Dec 31, 2010-Aug 31, 2026; 3,939 observations
Developer / validator	Cassie Wan / Sai Vakkalagadda

2. Methodology and conceptual soundness

For each day t, simple return r(t) = P(t)/P(t-1) - 1 and P&L(t) = $1,000,000 x r(t). VaR(t) is minus the 1% quantile of P&Ls on days t-250 through t-1. The forecast excludes the realized return on day t. The model is appropriate as a transparent empirical baseline for a linear SPY exposure; it does not impose a Normal return distribution.

The implementation uses linear quantile interpolation. With 250 observations, the zero-based rank is (250-1) x 0.01 = 2.49. Thus the quantile equals 0.51 times the third-worst P&L plus 0.49 times the fourth-worst P&L. This precise convention matters because so few observations determine the tail. It should be explicit in the model documentation. [2]

The main assumption is that the previous trading year represents next-day risk. Equal weights do not adapt directly to current volatility, so the model can lag a sudden shock and remain elevated after conditions normalize. Tail estimates depend on individual observations entering or leaving the window. The estimate cannot extrapolate beyond the historical window, and VaR does not measure losses beyond the threshold.

The 250-day window is the agreed project baseline, not evidence of an optimal choice for SPY. Constant exposure avoids position-size drift. Price-only returns exclude dividends and must remain consistent across models. The original notes discuss adjusted prices, whereas the adopted implementation and development summary explicitly use unadjusted prices; this scope choice is recorded rather than silently changed.

3. Data and implementation review

An independent implementation reads the pinned raw data, explicitly enforces the Jan 2010-Aug 31, 2026 range, calculates simple returns and dollar P&L, and uses a lagged pandas rolling quantile. It does not call the developer model to obtain its validation estimates. All 3,939 equal-weighted forecasts agree within $0.000001. The age-weighted series and both benchmark model series also reproduce within that tolerance.

Checks confirm positive finite prices, unique dates in the project sample, no records after the freeze date, finite output fields, positive VaR, identical forecast calendars and matching returns/P&L across all models. The first 250 returns are used only for initialization. Changing day t and future P&Ls leaves the independent Historical forecast through t unchanged; source inspection confirms the developer uses the same exclusive t-250:t window.

Forecast date	Window end	Independent VaR	Reported VaR
2010-12-31	2010-12-30	$32,058.95	$32,058.95
2011-08-08	2011-08-05	$24,035.13	$24,035.13
2020-03-16	2020-03-13	$46,868.40	$46,868.40
2022-06-13	2022-06-10	$33,815.32	$33,815.32
2025-04-04	2025-04-03	$27,904.53	$27,904.53

The five checks include the first forecast, 2011 stress, the March 2020 crash, 2022 tightening and April 2025. The CSV includes window starts, third/fourth order statistics and unrounded errors. This verifies calculations separately from assessing predictive performance.

4. Implementation findings and actions

Finding	Impact and recommended action
Stale README	README still says log returns and missing Historical documentation. Current code and development summary use simple returns, and the summary exists. Update the README.
Hard-coded look-ahead check	Developer appends True for this check. The actual window is correctly lagged; replace the constant with an executable future-data perturbation check.
Freeze-date guard	Developer filters the start date only. Current data ends on the agreed cutoff, so this does not change present results. Add an explicit end-date filter/assertion.
Duplicate handling	Developer keeps the highest price for each date. No duplicate dates occur in the evaluated sample. Prefer rejecting conflicting duplicates or reconciling with the source.
Quantile documentation	State the linear 1% interpolation convention and the third/fourth order statistics. For age-weighting, document cumulative-midpoint interpolation.

Age-weighted detail: the code normalizes weights proportional to 0.97^age, where the newest observation has age zero. The document describes the newest observation as i=1 while displaying a lambda^i formula with a zero-based normalizer. This indexing inconsistency should be corrected; it does not affect the reproduced code output.

Baseline commit: bd49450e23f0a4d64703ba90298ceb4cc8e8ccf5. SHA-256 input hashes, package versions and assertion results are in results/provenance.json and results/implementation_checks.csv. The baseline files were preserved unchanged.

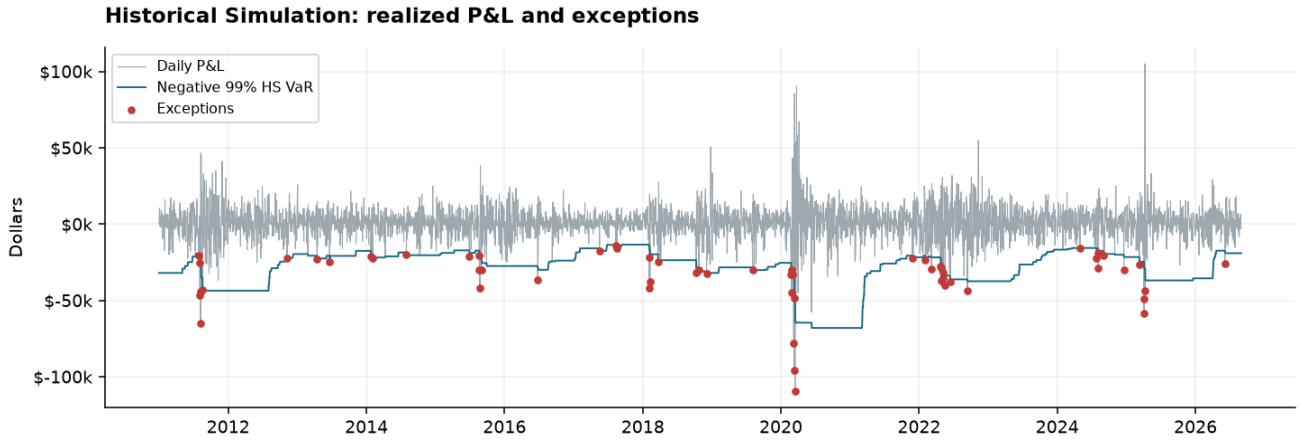
5. Backtesting & Benchmarking: exception analysis

An exception is strictly P&L(t) < -VaR(t); equality is not an exception. Tests use every out-of-sample observation, with consecutive trading observations defining transitions. Expected exceptions are 0.01 x 3,939 = 39.39. Actual exceptions are 59 (1.4978%), about 50% more than expected.

Test	LR statistic	p-value	Decision at 5%
Kupiec POF	8.5538	0.003448	Reject 1% coverage
Christoffersen independence	18.0824	0.00002116	Reject independence

Kupiec compares the Bernoulli likelihood at p=0.01 with the likelihood at the observed exception rate. Christoffersen compares a pooled transition probability with separate probabilities following a non-exception and an exception. Each LR statistic uses a chi-square distribution with one degree of freedom. Zero counts use the limiting convention 0 log 0 = 0; an unobserved preceding state is flagged as unidentified. [3,4]

Transition counts are N00=3,827, N01=52, N10=52 and N11=7 (total 3,938). The estimated next-day exception probability is 1.34% following a non-exception, versus 11.86% following an exception. This is evidence of clustering. An exact binomial coverage diagnostic also rejects (p=0.002943), although its independence assumption is itself challenged by these results.



Year	Actual / expected	Year	Actual / expected	Year	Actual / expected
2010*	0 / 0.01	2016	1 / 2.52	2022	10 / 2.51
2011	6 / 2.52	2017	3 / 2.51	2023	0 / 2.50
2012	1 / 2.50	2018	7 / 2.51	2024	6 / 2.52
2013	2 / 2.52	2019	1 / 2.52	2025	4 / 2.50
2014	3 / 2.52	2020	8 / 2.53	2026*	1 / 1.66
2015	5 / 2.52	2021	1 / 2.52		

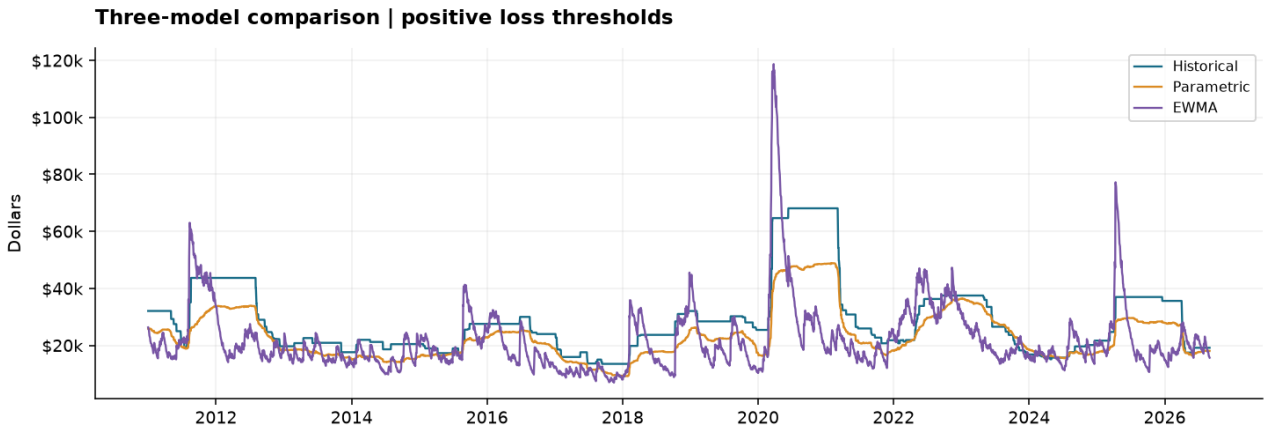
*Partial years. Annual counts are descriptive, not separate pass/fail tests: few exceptions are expected per year. The largest count is 10 in 2022, followed by 8 in 2020 and 7 in 2018. A non-rejection would not prove model accuracy; rejection requires investigation, not an automatic final rating.

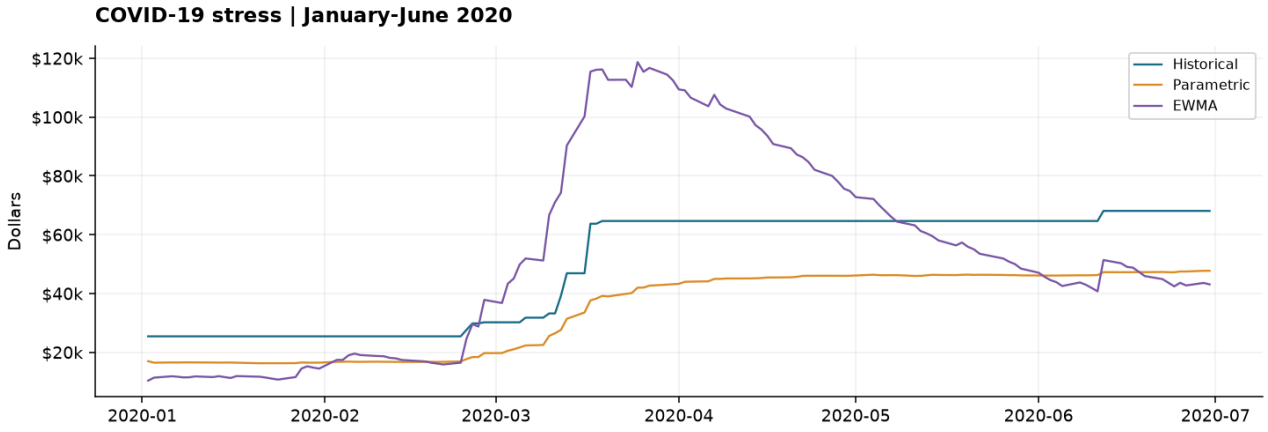
6. Benchmark comparison

All models use the same 3,939 dates and independently verified P&L. Historical has the highest average VaR and the fewest exceptions of the three primary models. This does not establish superiority: larger thresholds reduce exceptions at the cost of greater conservatism, and Historical still rejects both statistical tests.

Model	Avg VaR	Max VaR	Exceptions	Rate	Kupiec p	Indep. p
Historical	$28,603	$68,076	59	1.50%	0.00345	2.12e-05
Parametric	$23,188	$48,838	106	2.69%	1.15e-18	2.36e-05
EWMA	$21,915	$118,677	91	2.31%	1.65e-12	0.00584
Historical AW	$25,663	$109,424	73	1.85%	1.5e-06	0.0131

Historical AW is a supplementary age-weighted variant (lambda=0.97), not a replacement for the assigned baseline. It produces 73 exceptions and rejects both tests despite responding more strongly during stress.





In the specified February-June 2020 stress window (104 days), Historical has 8 exceptions, Parametric 13, EWMA 7 and age-weighted Historical 5. EWMA peaks at $118,677 on March 25, while equal-weighted Historical reaches its window peak of $68,076 on June 12. The chart shows a sharper EWMA response and a prolonged Historical plateau; peak dates alone are not a general response-speed statistic.

Historical records 6 exceptions in July-October 2011, 10 during 2022 and 3 in April 2025. These pre-specified calendar windows are descriptive stress diagnostics, not independent hypothesis tests. Full dates, counts and peak values are in stress_comparison.csv.

7. Sensitivity extension and preliminary findings

Sensitivity uses a common evaluation period of Dec 27, 2012-Aug 31, 2026 (3,439 days), allowing 750 prior returns for every variant. These results must not be compared directly with full-period counts of 59. Confidence levels are 95%, 97.5% and 99%; windows are 250, 500 and 750 days. This is exploratory analysis, not parameter selection on a holdout sample.

Window	Confidence	Exceptions / expected	Kupiec p	Indep. p
250	95.0%	177 / 171.95	0.694	8.02e-07
250	97.5%	113 / 85.98	0.00483	0.00131
250	99.0%	52 / 34.39	0.00502	8.77e-06
500	95.0%	168 / 171.95	0.756	1.44e-06
500	97.5%	104 / 85.98	0.0565	0.000294
500	99.0%	52 / 34.39	0.00502	6.24e-07
750	95.0%	168 / 171.95	0.756	5.36e-06
750	97.5%	86 / 85.98	0.998	4.69e-05
750	99.0%	39 / 34.39	0.439	2.92e-06

At 99%, extending the window to 750 days reduces exceptions from 52 to 39 and coverage is not rejected (p=0.439), but independence remains rejected. All nine variants reject independence. A longer window therefore does not resolve clustering and is not established as optimal. Multiple exploratory tests are unadjusted and require cautious interpretation.

Changing only the 250-day quantile convention yields 54 exceptions for lower/nearest, 59 for linear and 68 for higher. All reject coverage and independence at 5%. This supports the need to document interpolation, while showing the central finding is not confined to the linear convention.

Priority / finding	Recommended next step
High: coverage shortfall	Investigate loss episodes and quantify exceedance severity. Retain the fixed baseline for reporting; assess any revision out of sample.
High: clustered exceptions	Assess volatility-aware historical challengers or filtered historical simulation, with separate calibration and evaluation.
Medium: tail instability	Review window and quantile sensitivity; the 99% estimate depends on sparse tail observations.
Medium: documentation/control gaps	Correct README, quantile and age-index descriptions; automate data-cutoff and look-ahead checks.

Limitations: nominal p-values use asymptotic reference distributions; overlapping windows and changing regimes complicate interpretation. The frozen dataset was internally checked but Bloomberg source prices were not independently authenticated. Results describe a constant, price-only SPY exposure. No final effectiveness rating is assigned.

Sources and reproducibility

[1] Team repository at commit bd49450e23f0: Historical script, development summary, raw SPY data and model outputs; General Notes.pdf, pp. 22-31. Repository: https://github.com/sohamjoshi1998/fim500-var-models

[2] NumPy percentile documentation: https://numpy.org/doc/stable/reference/generated/numpy.percentile.html

[3] MathWorks Kupiec POF: https://www.mathworks.com/help/risk/varbacktest.pof.html [4] MathWorks Christoffersen independence: https://www.mathworks.com/help/risk/varbacktest.cci.html. The published independence LR example is reproduced by an assertion in the notebook. Accessed October 23, 2026.